Group 2

AI v.s. Human Generated Text Analysis

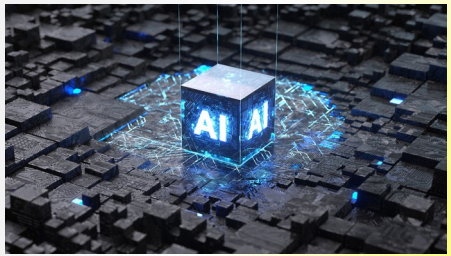
Van (Betty) Do, Zhihao (Vincent) Feng, Hannan Harahap, Izumi Kin, Tiffany Leong, Josiah Ohara, Binbo Xu.

Project Context

The Rise of AI-Generated Text

Rapid Advancement in LLM & Generative AI

With the advancement of technology, AI can now produce human-like essays, articles, and conversations at scale



Challenges Across Sectors



There are many industries that face the challenge stemming from AI capabilities

- **Education / academic institutions** face Growth in **AI-assisted plagiarism and academic dishonesty**, difficulty in distinguishing original student work from AI-generated content.
- **Wider range of professional industries** such as media faces **erosion of of trust** in online content due to surge in **AI-generated fake news and misinformation**
- **General public** is now at **higher risk of manipulation and misinformation** spread due to the blurred line between authentic and synthetic communication

Project Dataset Overview

Data Sources:

We have **merged 3 different dataset** sourced from ArguGPT (HuggingFace). LLM-Generated Essays (Kaggle), and DAIGT External Train Dataset (Kaggle), in order to have sufficient input of both AI generated and human generated text data in model training. We have **standardized the variables**.

Dataset Structure & Key Variables:

Size: 2,750 samples
Input Feature: *text*
Target Label: *generated* (0=human, 1=AI)
Class Distribution: 50 / 50 human and AI split

Dataset Snapshot:

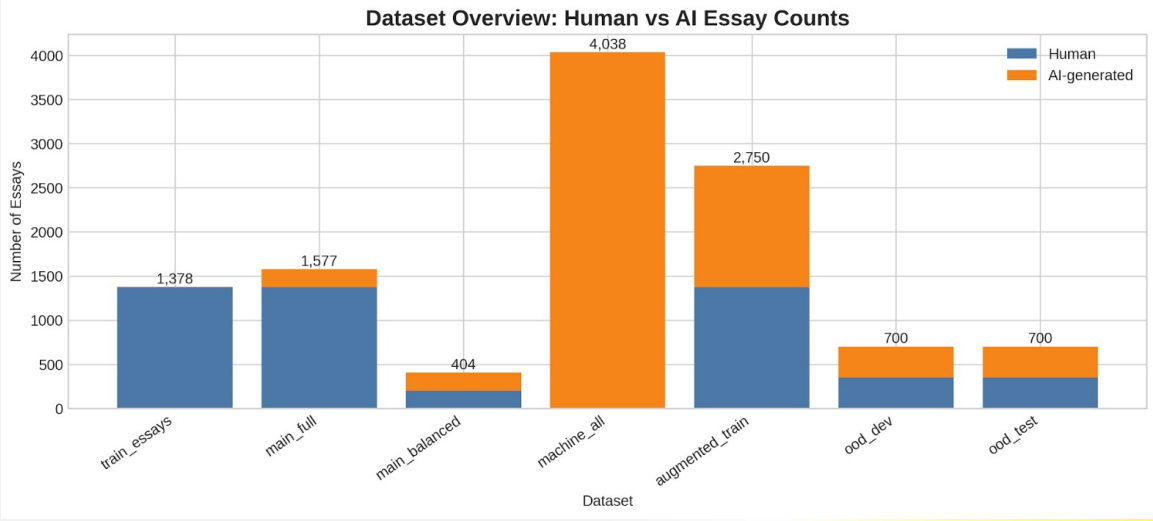
id	text	prompt_id_raw	generated	source_dataset	prompt_text	model_name	source_split	word_count	char_count
wecci_1076	There is no one answ One reason why it m Another reason why i	WECCL-08	1	argugpt	In the western world,	text-babbage-001	train	202	1130
afe7cdd4	Dear State Senator, First of all, an examp Secondly, this is not Third, according to t Conclusively, the El	1	0	kaggle_comp		unknown	train	459	2622
wecci_876	Famous people are c On one hand, it is ur On the other hand, it Ultimately, I believe	WECCL-10	1	argugpt	Some people think tl	text-curie-001	train	245	1391
	It is certainly true tha One reason is that yo Another reason is tha Finally, young peopl								

Dataset Characteristics:

The dataset consists of structured, prompt-driven text with a balanced mix of human and AI-generated samples, enabling controlled comparison across similar contexts. Its diversity in style and multi-source composition supports the analysis of features such as coherence, variability, and repetition, improving the model's ability to generalize to real-world scenarios.

Summary Statistics & Initial Findings

What does the data tell us?



Importance of Balanced Data

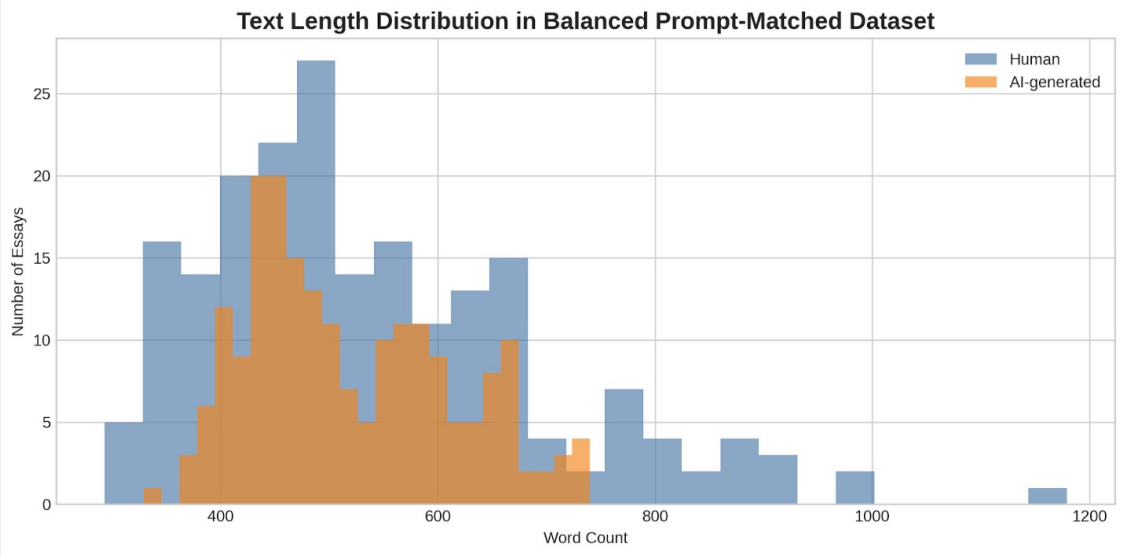
The main_balanced represents the near-equal split between human and AI-generated content. Detection models trained on **skewed datasets risk being biased**, **potentially misclassifying** human writing as AI-generated more often than they should.

Human-Written Essays Remain the Minority in Training Data

Across most datasets, human essays are consistently outnumbered by AI-generated ones. This **imbalance in training data** raises important questions about **whether detection models can remain fair and accurate**.

Summary Statistics & Initial Findings Cont'd.

What does the data tell us?



AI-Generated Essays has a Narrow Word Range

AI-generated essays are heavily concentrated between **400–700 words**, with very few venturing beyond that range. This suggests AI tends to produce consistently **'safe' length** responses, which could itself become a detectable pattern for detection models.

Human Writers Show Far Greater Length Variability

Unlike AI, human essays spread across a much wider range, from under 300 words all the way past 1,200, reflecting the **natural inconsistency** in how people write

Key Questions

Main Question

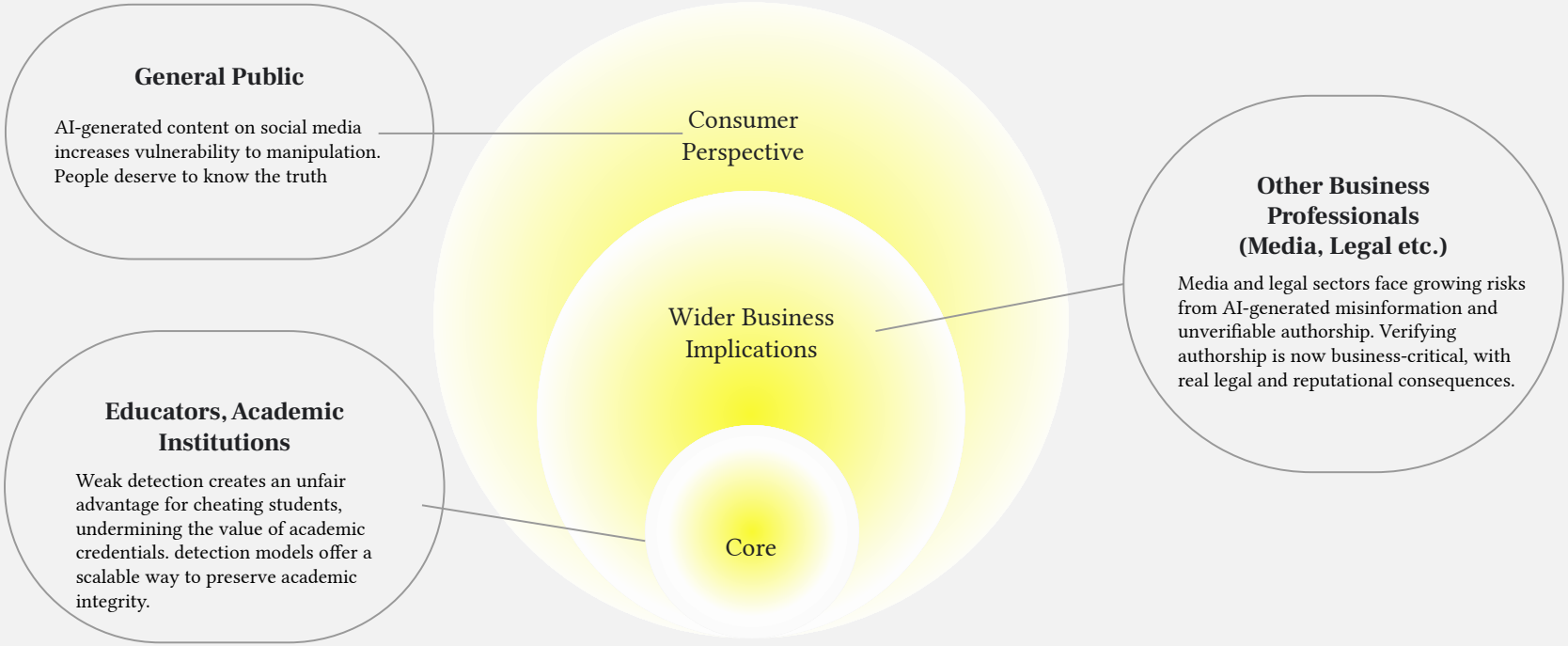


How can we accurately identify whether a piece of text is AI-generated or human-written?

Sub-questions

1. What distinguishes AI-generated text from human writing?
2. Which features are most useful for detection?
3. What models or approaches can effectively perform this classification?
4. How do we evaluate detection performance?
5. What are the limitations and risks of AI detection systems?

Target Audience & Motivation



Overview of Analysis Plan - Decomposed Problem Statement

1

What linguistic and structural features differ between AI-generated and human-written essays?

This focuses on identifying *what to measure* — things like sentence length consistency, vocabulary diversity, punctuation patterns, and word predictability

2

How accurately can models classify essays when trained on balanced vs. imbalanced datasets?

This focuses on *model performance* — directly tied to what your charts showed about dataset composition.

3

How well do detection models generalize to essays they were not trained on (out-of-distribution text)?

This focuses on *real-world reliability*.
A model that performs well on training data but fails on unseen essays would not be practical for educators or institutions trying to detect AI use in the wild.

Methodology

We compared traditional and deep learning approaches to identify and interpret patterns that distinguish AI-generated text from human writing

Stage	Method / Model	Purpose
Feature Extraction	TF-IDF	Convert text into numerical features based on word importance
Modeling (Traditional ML)	Logistic Regression, Random Forest	Classify text using statistical patterns
Modeling (Deep Learning)	BERT (Bidirectional Encoder Representations from Transformers)	Capture contextual and semantic meaning in text
Model Interpretation	LIME (Local Interpretable Model-agnostic Explanations)	Explain model predictions and identify important features

Traditional models use TF-IDF features, while BERT leverages contextual embeddings directly for classification.

Advanced Analytics - Two Models, One Pipeline

	BERT (classification model)	LIME (Explainability model)
How it works	Reads text bidirectionally to deeply understand context, then classifies each essay as AI or human.	Perturbs words in the text and observes prediction changes to build a local explanation for each decision.
Output	Binary label + confidence score	Feature importance scores + highlighted words
Strength	High accuracy on complex, nuanced text	Makes findings transparent and auditable
Limitation	acts as a black box — verdict without reasoning	explains one prediction at a time — always paired with BERT

Note:
BERT detects. LIME explains. Together they deliver both accuracy and transparency, which is essential for stakeholders who need to justify every decision.

Advanced Analytics Cont'd.

BERT achieves ~97% accuracy in classifying AI vs human essays. LIME reveals the word-level patterns driving each prediction, making results both accurate and explainable

BERT Performance

Bidirectional transformer fine-tuned on essay datasets. Outperforms traditional models (TF-IDF, bag-of-words) which plateau at 80–85% accuracy. Chosen for its ability to capture deep contextual meaning across full text

LIME Explainability

Assigns word-level weights to explain individual BERT predictions. Words like "subject" (+0.0044) and "academic" (+0.0029) flagged as AI indicators. Chosen to open BERT's black box for stakeholders who need justifiable evidence

3 Key Metrics

BERT Classification Accuracy

97%

highest LIME word weight ("subject")

+0.0044

accuracy gap over traditional NLP baselines

~15%

Impact - beyond education

Educators

- Greater power to identify AI-generated essays, reports, and dissertations
- Ability to enforce academic integrity policies with more confidence
- Helps ensure students are genuinely developing critical thinking and writing skills

Journalist / Media

- Stronger ability to flag and remove AI-generated misinformation before it spreads
- Builds and maintains audience trust by guaranteeing authentic reporting
- Helps distinguish legitimate news from synthetic propaganda at scale

Legal

- Provides evidence in cases involving fraud, defamation, or copyright disputes tied to AI content
- Enables enforcement of AI disclosure laws that would otherwise be unverifiable
- Gives regulators a concrete tool to hold companies and individuals accountable

Social Media Users

- Better protection from AI-generated fake reviews, scams, and manipulative content
- More informed decision-making when consuming news, product reviews, or opinions
- Reduces the spread of synthetic personas used in influence campaigns

Key takeaway

BERT ✦ LIME achieved **97% accuracy** while **remaining transparent and explainable**; a pipeline built for real-world stakeholder trust

What we found

BERT outperformed traditional NLP by ~15%. LIME exposed that AI essays rely on predictable, formal vocabulary. Dataset imbalance remains the biggest risk to model fairness.

Where do we go next

Test generalization on out-of-distribution essays. Explore whether detection holds as AI writing tools evolve. Expand beyond academic essays to social media and news content.

Challenge faced

What we faced, and how we addressed them

Choosing the right data source	We faced difficulties figuring out what were the appropriate dataset to use. **We have taken a step back to better understand our problem statement to pinpoint and identify what kind of data we need**
Inconsistent data structure / labels	Inconsistency in columns and structure, length of text data made merging of multiple dataset difficult. **We have taken time to brainstorm standardization**
Overfitting of models	Our models indicated the sign of overfitting. **We have further investigated the dataset we used and adjusted the modeling approach**
Models giving different type of output	Our models learned what sources the text data was from instead of identifying the words that differentiated AI vs human. **We have further investigated into our modeling approach and made adjustments**

Any questions? Ask away!

Contact us at name@example.com